Determine the derivative!:

$$f(x) = \ln \sqrt[3]{\frac{x}{2x^2+3}}$$

$$f(x) = \frac{1}{3}\ln\left(\frac{x}{2x^2+3}\right) = \frac{1}{3}\left(\ln x - \ln(2x^2+3)\right)$$

$$f'(x) = \frac{1}{3}\left(\frac{1}{x} - \frac{4x}{2x^2+3}\right)$$

$$f'(x) = \frac{1}{3}\left(\frac{2x^2+3}{x(2x^2+3)} - \frac{x(4x)}{x(2x^2+3)}\right)$$

$$f'(x) = \frac{1}{3}\left(\frac{2x^2+3-4x^2}{x(2x^2+3)}\right)$$

$$\boxed{f'(x) = \frac{-2x^2+3}{3x(2x^2+3)}}$$

Day 5